IPL 2024 REPORT

BY

SURAJ PAL

GUIDEDBY- MR. SAMIR WARSOLKARSIR

INDIAN
PREMIER
LEAGUE
DELHI CAPITALS
KINGS
SUPER KINGS
ROYALS
CHALLENGERS
INDIANS
TITANS

INTRODUCTION:

The Indian Premier League (IPL) has established itself as one of the most popular and competitive cricket leagues in the world, drawing attention from millions of fans globally. The IPL 2024 season promises to be even more exciting, with new players, strategies, and teams all vying for the championship. Given the vast amount of data generated during each season, analyzing and understanding this data becomes essential for sports analysts, fans, and stakeholders.

This SQL project focuses on creating a comprehensive IPL 2024 report by utilizing various database management and SQL techniques. The goal is to analyze the performance metrics of teams and players, match statistics, and other key aspects of the season. The project will involve tasks such as querying data to extract relevant insights, organizing large datasets, and visualizing the results to provide a clear understanding of team and player performance.

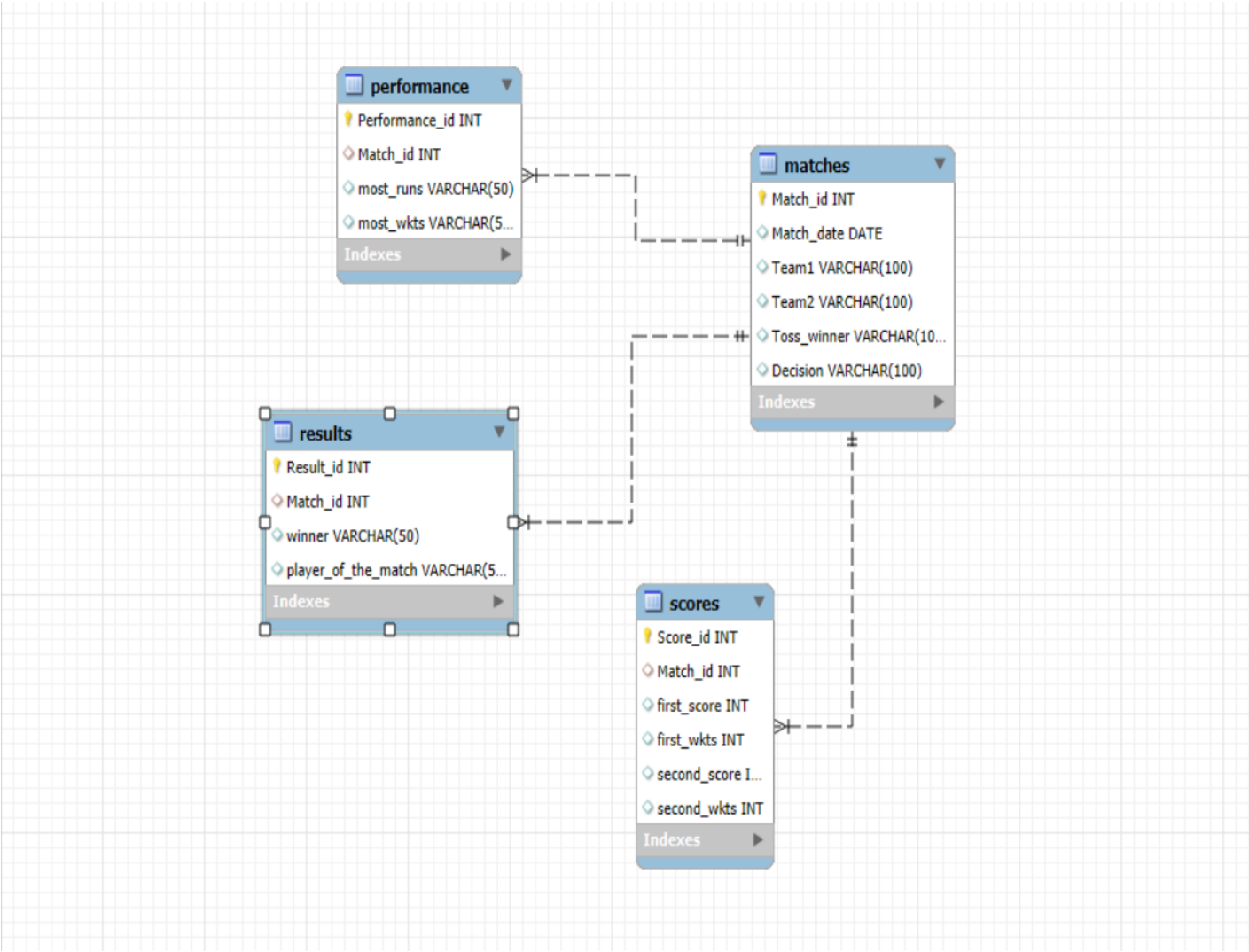
ABSTRACT:

The Indian Premier League (IPL) is one of the most celebrated cricket tournaments, drawing significant attention globally due to its competitive nature and star-studded lineup. This SQL project aims to create a detailed analytical report on the IPL 2024 season by employing structured query language (SQL) to manage, query, and analyze large datasets. The project involves extracting and processing key data points related to player performances, match statistics, team rankings, and tournament outcomes.

The primary focus is to deliver insights on critical aspects of the season, including top-performing players, win/loss ratios, batting and bowling averages, and trends in team performance. SQL queries will be designed to derive these insights from complex datasets, while also visualizing patterns that reveal the underlying dynamics of the tournament.

The IPL 2024 report aims to provide a data-driven narrative that enhances understanding of player and team performance throughout the season. This analysis is beneficial for sports analysts, enthusiasts, and stakeholders interested in deriving meaningful conclusions from large sports datasets. Through SQL's power to manipulate and analyze data efficiently, this project demonstrates the importance of database management in sports analytics.

ER-DIAGRAM:-


performance
Performance_id INT
Match_id INT
most_runs VARCHAR(50)
most_wkts VARCHAR(5...
Indexes
matches
Match_id INT
Match_date DATE
Team1 VARCHAR(100)
Team2 VARCHAR(100)
Toss_winner VARCHAR(10...
Decision VARCHAR(100)
Indexes
results
Result_id INT
Match_id INT
winner VARCHAR(50)
player_of_the_match VARCHAR(5...
Indexes
scores
Score_id INT
Match_id INT
first_score INT
first_wkts INT
second_score I...
second_wkts INT
Indexes

COMMANDS :-

Create Database:-

CREATE DATABASE IPL2024;

Select Database:-

USE IPL2024;

Create table named Matches:

```
CREATE TABLE Matches
(Match_id INT PRIMARY KEY,
Match_date DATE ,
Team1 varchar(100),
Team2 VARCHAR(100),
Toss_winner VARCHAR(100),
Decision VARCHAR(100));
```

Create table named Scores:-

```
CREATE TABLE Scores (
Score_id INT PRIMARY KEY,
Match_id INT,
first_score INT,
first_wkts INT,
second_score INT,
second_wkts INT,
FOREIGN KEY (Match_id) REFERENCES Matches(Match_id));
```

Create table named Results:-

```
CREATE TABLE Results (
Result_id INT PRIMARY KEY,
Match_id INT,
winner VARCHAR(50)
player_of_the_match VARCHAR(50),
FOREIGN KEY (match_id) REFERENCES Matches(match_id));
```

Create table named Performance:-

```
CREATE TABLE Performance (
Performance_id INT PRIMARY KEY,
Match_id INT,
most_runs VARCHAR(50),
most_wkts VARCHAR(50),
FOREIGN KEY (Match_id) REFERENCES Matches(Match_id));
```

Table Descriptions:

1.Matches:-

	Field	Type	Null	Key	Default	Extra
▶	Match_id	int	NO	PRI	NULL	
	date	text	YES		NULL	
	team1	text	YES		NULL	
	team2	text	YES		NULL	
	toss_winner	text	YES		NULL	
	decision	text	YES		NULL	

2.Scores Table:-

	Field	Type	Null	Key	Default	Extra
▶	Score_id	int	NO	PRI	NULL	
	Match_id	int	YES	MUL	NULL	
	first_score	int	YES		NULL	
	first_wkts	int	YES		NULL	
	second_score	int	YES		NULL	
	second_wkts	int	YES		NULL	

3.Results Table:-

	Field	Type	Null	Key	Default	Extra
▶	Result_id	int	NO	PRI	NULL	
	Match_id	int	YES	MUL	NULL	
	winner	varchar(50)	YES		NULL	
	player_of_the_match	varchar(50)	YES		NULL	

4.Performance table:-

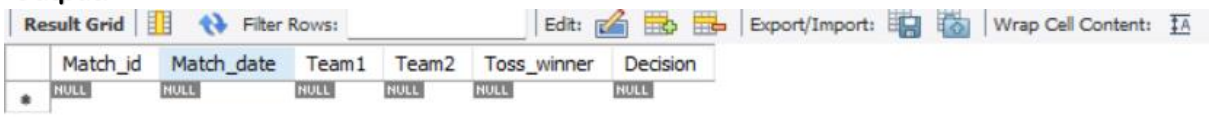
	Field	Type	Null	Key	Default	Extra
▶	Performance_id	int	NO	PRI	NULL	
	Match_id	int	YES	MUL	NULL	
	most_runs	varchar(50)	YES		NULL	
	most_wkts	varchar(50)	YES		NULL	

Import Data Into Tables:-

Import Data In Matches Table

Step 1: SELECT * FROM Matches;

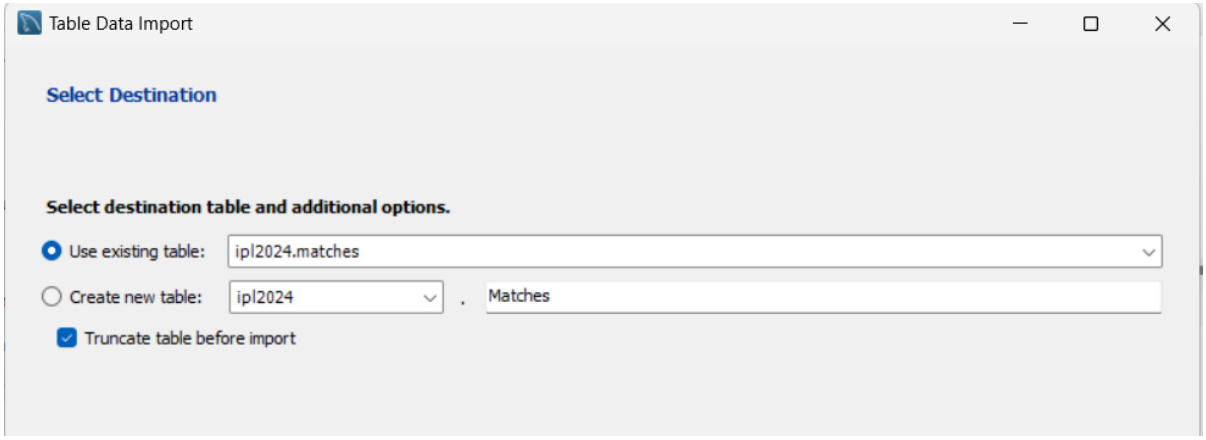
Output:



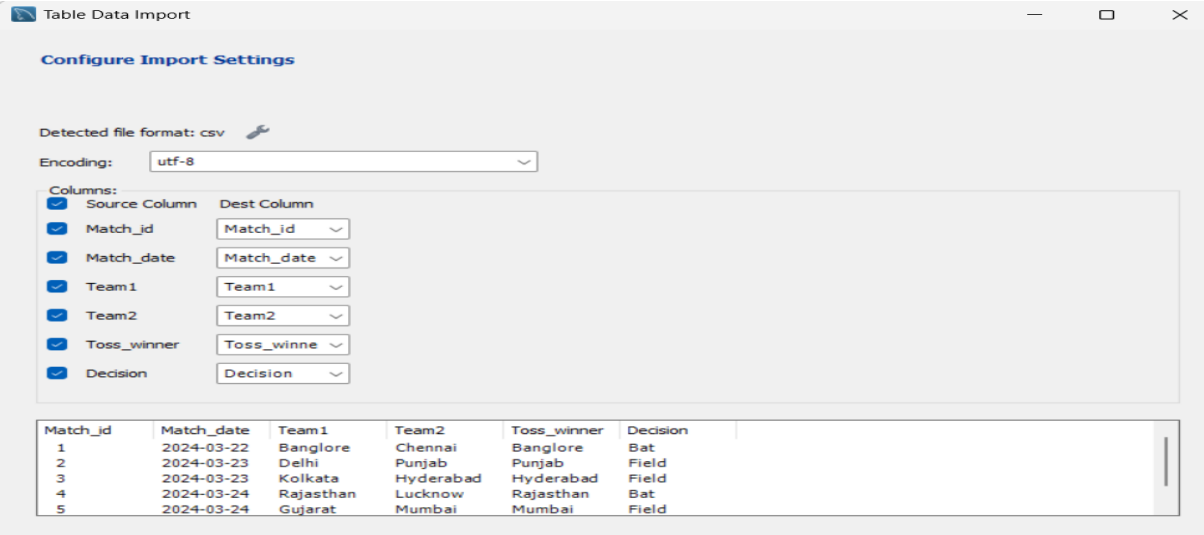
Step 2: Right click on Import data as show in the below picture

Step 2: select csv file of matches table

Step 3: Select destination table as show in the below picture and click on the next button.



Step 4: Configure Import setting -



Step 5: click next then click on the Import data

Step 6: SELECT * FROM Matches;

Output:

Match_id	Match_date	Team1	Team2	Toss_winner	Decision
1	2024-03-22	Banglore	Chennai	Banglore	Bat
2	2024-03-23	Delhi	Punjab	Punjab	Field
3	2024-03-23	Kolkata	Hyderabad	Hyderabad	Field
4	2024-03-24	Rajasthan	Lucknow	Rajasthan	Bat
5	2024-03-24	Gujarat	Mumbai	Mumbai	Field
6	2024-03-25	Punjab	Banglore	Banglore	Field
7	2024-03-26	Chennai	Gujarat	Gujarat	Field
8	2024-03-27	Hyderabad	Mumbai	Mumbai	Field
9	2024-03-28	Rajasthan	Delhi	Delhi	Field
10	2024-03-29	Banglore	Kolkata	Kolkata	Field
11	2024-03-30	Lucknow	Punjab	Lucknow	Bat
12	2024-03-31	Hyderabad	Gujarat	Hyderabad	Bat
13	2024-03-31	Delhi	Chennai	Delhi	Bat

- **Follow the same steps for the remaining tables**
- **Show Scores Table data:**

Query: SELECT * FROM Scores;

Output:

Score_id	Match_id	first_score	first_wkts	second_score	second_wkts
1	1	173	6	176	4
2	2	174	9	177	6
3	3	208	7	204	7
4	4	193	4	173	6
5	5	168	6	162	9
6	6	176	6	178	6
7	7	206	6	143	8
8	8	277	3	246	5
9	9	185	5	173	5
10	10	182	6	186	3
11	11	199	8	178	5
12	12	162	8	168	3
13	13	191	5	171	6

- **Show Results Table data:**

Query: SELECT * FROM Scores;

Output:

Result_id	Match_id	winner	player_of_the_match
1	1	Chennai	Mustafizur Rahman
2	2	Punjab	Sam Curran
3	3	Kolkata	Andre Russell
4	4	Rajasthan	Sanju Samson
5	5	Gujarat	Sai Sudharsan
6	6	Banglore	Virat Kohli
7	7	Chennai	Shivam Dube
8	8	Hyderabad	Abhishek Sharma
9	9	Rajasthan	Riyan Parag
10	10	Kolkata	Sunil Narine
11	11	Lucknow	Mayank Yadav
12	12	Gujarat	Mohit Sharma
13	13	Delhi	Khaleel Ahmed

- **Show Performance Table data:**

  Query: SELECT * FROM Performance;

Output:

Performance_id	Match_id	most_runs	most_wkts
1	1	Anuj Rawat	Mustafizur Rahman
2	2	Sam Curran	Kuldeep Yadav
3	3	Andre Russell	T Natarajan
4	4	Sanju Samson	Trent Boult
5	5	Dewald Brevis	Jasprit Bumrah
6	6	Virat Kohli	Harpreet Brar
7	7	Shivam Dube	Tushar Deshpande
8	8	Heinrich Klaasen	Pat Cummins
9	9	Riyan Parag	Yuzvendra Chahal
10	10	Virat Kohli	Andre Russell
11	11	Shikhar Dhawan	Mayank Yadav
12	12	Sai Sudharsan	Mohit Sharma
13	13	Jos Buttler	Mukesh Kumar

- **Show Matches Table data:**

  SELECT * FROM Matches;

Match_id	Match_date	Team1	Team2	Toss_winner	Decision
1	2024-03-22	Banglore	Chennai	Banglore	Bat
2	2024-03-23	Delhi	Punjab	Punjab	Field
3	2024-03-23	Kolkata	Hyderabad	Hyderabad	Field
4	2024-03-24	Rajasthan	Lucknow	Rajasthan	Bat
5	2024-03-24	Gujarat	Mumbai	Mumbai	Field
6	2024-03-25	Punjab	Banglore	Banglore	Field
7	2024-03-26	Chennai	Gujarat	Gujarat	Field
8	2024-03-27	Hyderabad	Mumbai	Mumbai	Field
9	2024-03-28	Rajasthan	Delhi	Delhi	Field
10	2024-03-29	Banglore	Kolkata	Kolkata	Field
11	2024-03-30	Lucknow	Punjab	Lucknow	Bat
12	2024-03-31	Hyderabad	Gujarat	Hyderabad	Bat
13	2024-03-31	Delhi	Chennai	Delhi	Bat

1. INSERT Query

Q.Insert given values into table Matches . Match_id=75, Match_date=2024-05-28, Team1=Banglore,Team2=Chennai ,Toss_winner= Banglore ,Decision=Bat

Query: INSERT INTO Matches (Match_id, Match_date, Team1, Team2, Toss_winner, Decision)VALUES (75, '2024-05-28', 'Banglore', 'Chennai', 'Banglore', 'Bat');

Output:-

18 00:30:16 INSERT INTO Matches (Match_id, Match_date, Team1, Team2, Toss_winner, Decision) VALUES (75, '2024-... 1 row(s) affected

74	2024-05-26	Hyderabad	Kolkata	Hyderabad	Bat
75	2024-05-28	Banglore	Chennai	Banglore	Bat
NULL	NULL	NULL	NULL	NULL	NULL

2. SELECT Query:-

Q. Retrive The details of the matches where the decision after winning the toss was 'field'.

Query: SELECT * FROM matches WHERE Decision='Field';

Output:

	Match_id	Match_date	Team1	Team2	Toss_winner	Decision
▶	2	2024-03-23	Delhi	Punjab	Punjab	Field
	3	2024-03-23	Kolkata	Hyderabad	Hyderabad	Field
	5	2024-03-24	Gujarat	Mumbai	Mumbai	Field
	6	2024-03-25	Punjab	Banglore	Banglore	Field
	7	2024-03-26	Chennai	Gujarat	Gujarat	Field
	8	2024-03-27	Hyderabad	Mumbai	Mumbai	Field
	9	2024-03-28	Rajasthan	Delhi	Delhi	Field
	10	2024-03-29	Banglore	Kolkata	Kolkata	Field
	14	2024-04-01	Mumbai	Rajasthan	Rajasthan	Field
	15	2024-04-02	Lucknow	Banglore	Banglore	Field
	17	2024-04-04	Gujarat	Punjab	Punjab	Field

3. UPDATE Query:-

Q. How do you update the team1 for a match_id =75 in the Matches table to Decision'Bat' where the Toss_winner is 'team1' and the Match_date is '2024-05-27'

Query:- UPDATE Matches SET team1 = 'Mumbai', Decision = 'Bat',Toss_winner = ‘Mumbai',Match_date = '2024-09-10' WHERE Match_id=75;

SELECT * FROM Matches;

Output:-

25 01:07:56 UPDATE Matches SET team1 = 'Mumbai', Decision = 'Bat',Toss_winner = 'Mumbai',Match_date = '2024-09-10'... 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

75	2024-09-10	Mumbai	Chennai	Mumbai	Bat
NULL	NULL	NULL	NULL	NULL	NULL

4. DELETE Query:-

Q. Delete the match record from the Matches table where the Match_id is 75

Query:- DELETE FROM Matches WHERE Match_id = 75;

OUTPUT:-

27	01:18:18	DELETE FROM Matches WHERE Match_id = 75	1 row(s) affected

5. Group by Clause:-

Q: Find the number of matches each team has played

Query:-
```
SELECT Team1 AS Team, COUNT(Match_id) AS Matches_Played
FROM Matches
GROUP BY Team1
UNION
SELECT Team2 AS Team, COUNT(Match_id) AS Matches_Played
FROM Matches
GROUP BY Team2;
```

Output:-

Team	Matches_Played
Banglore	9
Delhi	7
Kolkata	9
Rajasthan	6
Gujarat	7
Punjab	4
Chennai	8
Hyderabad	11
Lucknow	8
Mumbai	5
Chennai	6

6. Order by clause:-

Q: Find the top 5 highest-scoring matches based on the total score of both teams.

Query:- SELECT Match_id, (first_score + second_score) AS Total_Runs

FROM Scores

ORDER BY Total_Runs DESC LIMIT 5;

Output:-

Match_id	Total_Runs
42	523
8	523
43	504
35	465
30	449

7.Having clause :-

Q. Find players who have won "Player of the Match" more than once**.**

Query:-SELECT player_of_the_match, COUNT(result_id) AS Awards

FROM Results

GROUP BY player_of_the_match

HAVING COUNT(result_id) > 1;

Output:-

player_of_the_match	Awards
Sam Curran	2
Andre Russell	2
Sanju Samson	2
Virat Kohli	2
Abhishek Sharma	3
Sunil Narine	3
Mayank Yadav	2
Jos Buttler	2
Ravindra Jadeja	2
Jasprit Bumrah	2
Kuldeep Yadav	2

8.Using Distinct function :-

Q: Find all distinct teams that have participated in matches.

Query : SELECT DISTINCT Team1 AS Team FROM Matches

UNION SELECT DISTINCT Team2 AS Team

FROM Matches;

Output:

9.Using Like function:-

Q. Find all players who won "Player of the Match" and have a name starting with the letter "A".

Team
Banglore
Delhi
Kolkata
Rajasthan
Gujarat
Punjab
Chennai
Hyderabad
Lucknow
Mumbai

Query:- SELECT player_of_the_match, match_id

FROM Results

WHERE player_of_the_match LIKE 'A%';

Output:-

	player_of_the_match	match_id
▶	Andre Russell	3
	Abhishek Sharma	8
	Abhishek Sharma	18
	Andre Russell	36
	Abhishek Sharma	69

10. Using Between :-

Q. Find all matches that occurred between '2024-05-01' to '2024-05-31'

Result Grid | Filter Rows:

	Total_Matches
▶	74

Query:- SELECT Match_id, Match_date, Team1, Team2

FROM Matches

WHERE Match_date BETWEEN '2024-05-01' AND '2024-05-31';

Output:-

	Match_id	Match_date	Team1	Team2
▶	49	2024-05-01	Chennai	Punjab
	50	2024-05-02	Hyderabad	Rajasthan
	51	2024-05-03	Kolkata	Mumbai
	52	2024-05-04	Gujarat	Banglore
	53	2024-05-05	Chennai	Punjab
	54	2024-05-05	Kolkata	Lucknow
	55	2024-05-06	Hyderabad	Mumbai
	56	2024-05-07	Delhi	Rajasthan
	57	2024-05-08	Lucknow	Hyderabad
	58	2024-05-09	Banglore	Punjab
	59	2024-05-10	Gujarat	Chennai

11. Aggregate functions:-

A) Count

Q. Query to count the total number of matches played.

Query:- SELECT COUNT(Match_id) AS Total_Matches

FROM Matches;

Output:-

B). SUM:

Q. calculate the total runs scored by both teams across all matches.

Query:- SELECT SUM(first_score + second_score) AS Total_Runs

FROM Scores;

Output:-

Result Grid | Filter Rows:

	Total_Runs
▶	25871

C) AVG:

Q. What was the average score of teams batting first in IPL 2024

Query:- SELECT round(avg(first_score)) AS avg_first_inning_score

FROM Scores

JOIN Matches ON Scores.match_id = Matches.match_id;

Output:-

Result Grid | Filter Rows:

	avg_first_inning_score
▶	181

D)MIN:

Q. find the minimum score by Team 1 across all matches

Query:-SELECT MIN(first_score) AS Minimum_Team1_Score

FROM Scores

where first_score>1;

Output:-

	Minimum_Team1_Score
▶	89

E) MAX:

Q: What is the highest score made by any team in the first innings?

Query:- SELECT MAX(first_score) AS highest_first_innings_score FROM Scores;

Output:-

Result Grid | Filter Rows:

	highest_first_innings_score
▶	277

12. MULTIPLE CONDITIONS Query:-

Q: For each match, was the toss decision to bat or bowl, and did the toss-winning team win the match?

Query:- SELECT

```
    Matches.match_id,
    team1,
    team2,
    toss_winner,
```

```
decision,
winner
CASE
    WHEN toss_winner = winner THEN 'Toss winner won the match'
    ELSE 'Toss winner lost the match'
  END AS toss_outcome
FROM Matches
JOIN Results ON Matches.match_id = Results.match_id;
```

Output:-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

match_id	team1	team2	toss_winner	decision	winner	toss_outcome
1	Banglore	Chennai	Banglore	Bat	Chennai	Toss winner lost the match
2	Delhi	Punjab	Punjab	Field	Punjab	Toss winner won the match
3	Kolkata	Hyderabad	Hyderabad	Field	Kolkata	Toss winner lost the match
4	Rajasthan	Lucknow	Rajasthan	Bat	Rajasthan	Toss winner won the match
5	Gujarat	Mumbai	Mumbai	Field	Gujarat	Toss winner lost the match
6	Punjab	Banglore	Banglore	Field	Banglore	Toss winner won the match
7	Chennai	Gujarat	Gujarat	Field	Chennai	Toss winner lost the match
8	Hyderabad	Mumbai	Mumbai	Field	Hyderabad	Toss winner lost the match
9	Rajasthan	Delhi	Delhi	Field	Rajasthan	Toss winner lost the match
10	Banglore	Kolkata	Kolkata	Field	Kolkata	Toss winner won the match
11	Lucknow	Punjab	Lucknow	Bat	Lucknow	Toss winner won the match
12	Hyderabad	Gujarat	Hyderabad	Bat	Gujarat	Toss winner lost the match

13. Subquery :-

Q:Which team had the highest total score across all their winning matches in IPL 2024?

Query: SELECT winner, SUM(total_score) AS total_winning_score

```
FROM (
SELECT winner,
(CASE
    WHEN winner = team1 THEN first_score
    WHEN winner = team2 THEN second_score
```

```
END) AS total_score
FROM Results
JOIN Matches ON Results.match_id = Matches.match_id
JOIN Scores ON Matches.match_id = Scores.match_id
AS WinningScores
GROUP BY winner
ORDER BY total_winning_score DESC;
```

Output:-

Result Grid | Filter Rows:

winner	total_winning_score
Kolkata	2046
Hyderabad	1836
Rajasthan	1626
Banglore	1388
Delhi	1363
Lucknow	1295
Chennai	1253
Punjab	947
Gujarat	912
Mumbai	799
Abandoned	NULL

14.joins:-

Inner join :-

Q: Which teams won the most matches when they won the toss and decided to bat first?

Query:-

```
SELECT Matches.team1, COUNT(*) AS wins_batting_first
FROM Matches
INNER JOIN Results ON Matches.match_id = Results.match_id
```

```
WHERE Matches.toss_winner = Matches.team1 AND Matches.decision = 'bat'
AND Results.winner = Matches.team1
AND YEAR(Matches.match_date) = 2024
GROUP BY Matches.team1
ORDER BY wins_batting_first DESC;
```

Output:

Result Grid | Filter Rows:

team1	wins_batting_first
Lucknow	2
Rajasthan	1
Delhi	1
Kolkata	1
Banglore	1
Hyderabad	1

Left join :-

Q: What is the performance summary (total runs, total wickets) for the toss-winning team in each match?

Query:

```
SELECT Matches.toss_winner, SUM(Scores.first_score + Scores.second_score)
AS total_runs,
SUM(Scores.first_wkts + Scores.second_wkts) AS total_wickets
```

```
FROM Matches
LEFT JOIN Scores ON Matches.match_id = Scores.match_id
WHERE YEAR(Matches.match_date) = 2024
GROUP BY Matches.toss_winner
ORDER BY total_runs DESC;
```

Output:-

toss_winner	total_runs	total_wickets
Mumbai	3873	125
Rajasthan	3788	137
Punjab	3780	139
Lucknow	3176	101
Banglore	3011	98
Delhi	2442	96
Hyderabad	2370	81
Gujarat	1188	36
Kolkata	1129	35
Chennai	1114	35
No Toss	0	0

Conclusion:

This SQL project provided a comprehensive analysis of the 2024 Indian Premier League (IPL) season by leveraging database management techniques and Structured Query Language (SQL). The project effectively utilized various SQL queries to explore match statistics, player performances, and team dynamics throughout the tournament. A structured approach was taken to organize the data into key tables like *Matches*, *Scores*, *Results*, and *Performance*, allowing for efficient querying and analysis of the vast IPL dataset.

Key insights included:

- **Match Analysis:** Through SQL commands, detailed match data such as toss decisions, team performances, and results were analyzed. For example, queries provided

information about how toss decisions influenced match outcomes and which teams performed better under certain conditions.

- **Player Performances:** Player-specific statistics, such as the number of times a player won the "Player of the Match" award and the performance of top run scorers and wicket-takers, were analyzed to highlight star performers of the season.
- **Team Rankings and Trends:** The project explored team performance, focusing on aspects like win/loss ratios, highest-scoring matches, and teams with the most consistent performances. SQL functions like *GROUP BY*, *HAVING*, and *ORDER BY* were used to rank teams based on match outcomes and total scores.
- **Advanced Queries:** The use of joins and subqueries helped provide complex insights, such as identifying teams that consistently won when deciding to bat first after winning the toss, and examining trends in winning teams' overall performance.

This project demonstrated SQL's capability to manage, query, and analyze large-scale datasets efficiently, offering a clear and data-driven understanding of IPL 2024. The use of aggregation, conditional logic, and advanced functions provided deeper insights, making this project valuable for analysts, teams, and fans seeking to understand the dynamics of the tournament.

By successfully managing and extracting key data points, the project highlighted the importance of SQL in sports analytics, emphasizing how database management can be instrumental in evaluating and improving performance metrics in professional sports like IPL.

Thank

You